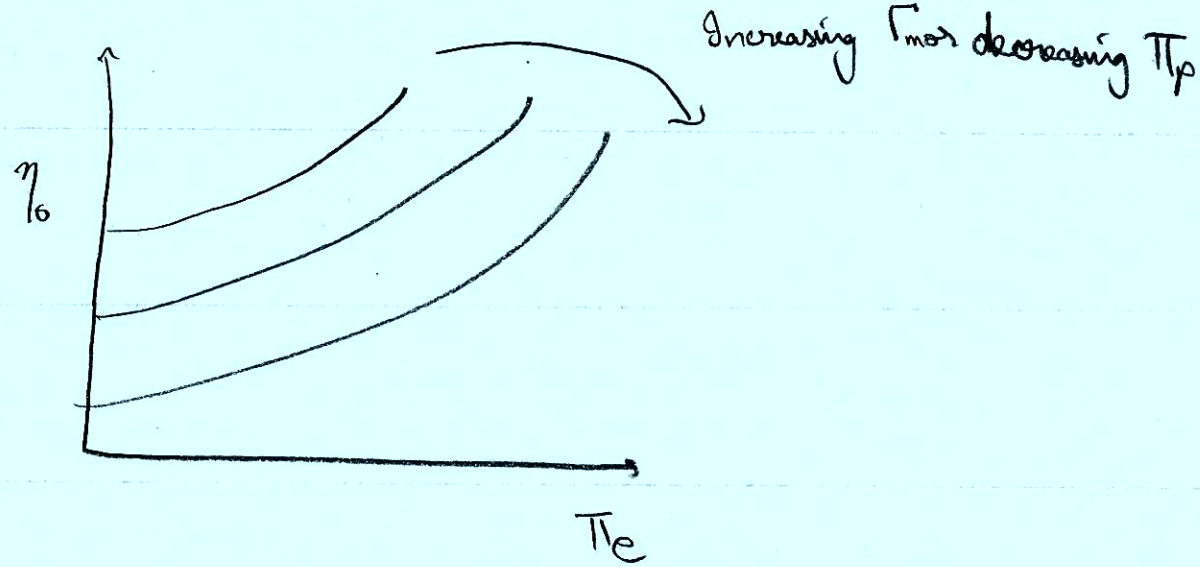
2/26/2019




- As $\Gamma_m$ decreases, need higher overall efficiency from propulsion system for a given empty mass fraction ($\Pi_e$)
- For a given propulsion system efficiency, as $\Pi_e$ increases, the payload decreases.
- The previous relations for transatmospheric flight were for a single stage.
- There is a large benefit to jettison an empty mass along the trajectory $\longrightarrow$ multistaging. (minimize $\Pi_e$)
- Will look @ 2 different config. of 2-stage to orbit vehicles (TSTO)
  - A) 1st stage is airbreathing, 2nd stage is also airbreathing
  - B) 1st stage is airbreathing, 2nd stage is rocket

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Review of rocket

For 2 stage vehicle.

Let subscript '1' refer to stage 1
" " '2' " " stage 2.

$$m_{i,1} = m_{p,1} + m_{e,1} + m_{f,1}$$

$$m_{i,2} = m_{p,2} + m_{e,2} + m_{f,2}$$

$$m_i = m_{i,1} + m_{i,2} \; ? \quad \text{WRONG!}$$

What is the payload of stage 1?

$$m_{p,1} = m_{i,2}$$

So, initial mass of vehicle is

$$m_i = m_{i,1} \quad (\text{b/c it includes } m_{i,2} \text{ in } m_{p,1})$$

True payload $m_p = m_{p,2}$

$$m_i = m_{p,2} + m_{f2} + m_{e2} + m_{f1} + m_{e1}$$

$$m_i = m_{i2}\left(\Pi_{p2} + \Pi_{f2} + \Pi_{e2}\right) + m_{i1}\left(\Pi_{f1} + \Pi_{e1}\right)$$

$$\frac{m_i}{m_p} = \Gamma_m\Big)_{\text{vehicle}} = \frac{m_{i1}}{m_{p2}} = \frac{m_{i1}}{m_{p1}} \cdot \frac{m_{p1}}{m_{p2}} = \frac{m_{i1}}{m_{p1}} \cdot \frac{m_{i2}}{m_{p2}} = \boxed{\Gamma_{m_1} \cdot \Gamma_{m_2} = \Gamma_{m_{\text{vehicle}}}}$$

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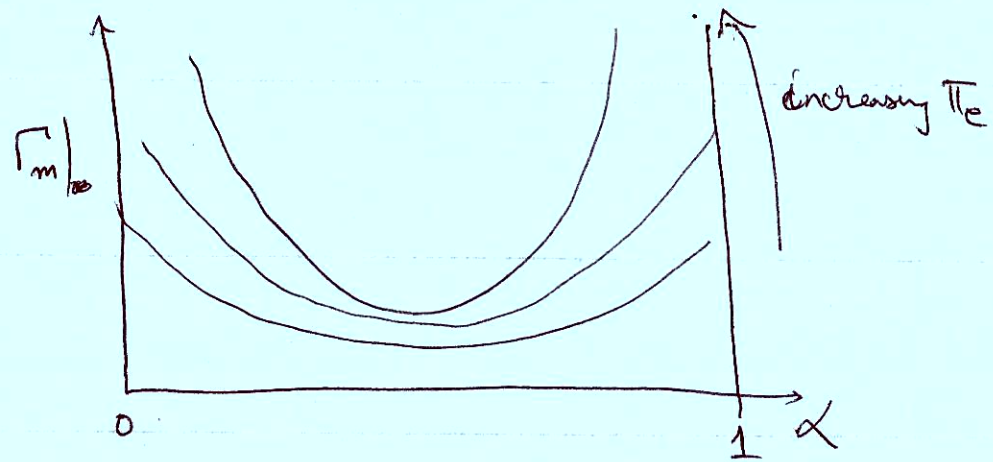
$\Rightarrow \Delta_1 = \Delta_2 = \Delta$ @ optimal

It can be further shown that we want each stage to do the same amount of work.

So, to ~~maximize~~ minimize $\Gamma_m$, we want $\alpha = 0.5$.

For optimal performance with a TSTO, airbreathing engines

Case A $\rightarrow$ $$\Gamma_m\Big|_{opt} = \left(\exp\left(-\frac{\Delta}{2}\right) - \Pi_e\right)^{-2}$$




As $\Gamma_e$ ~~increase~~ decreases $\Rightarrow$ $\Gamma_m$ decreases & become a very strong $f(\alpha)$

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For a rocket, we can define $I_{sp}$

$$I_{sp} = \frac{\int \tau \, dt}{\int \dot{m}_p g \, dt}$$

propellant for rocket

For a const $\tau$ & $\dot{m}_p$

$$I_{sp} = \frac{\tau}{\underbrace{\dot{m}_p g_0}}$$

propellant weight flux.

$I_{sp}$ – units of seconds

– want to maximize.

For cruise.

$$\Pi_f = 1 - \exp(-BR)$$

Use $\eta_0 = \dfrac{\tau V_0}{\dot{m}_f H_{prop}}$ ~~& rate of~~

Use $I_{sp} = \dfrac{\tau}{\dot{m}_f g_0}$ ($\dot{m}_f = \dot{m}_p$ in rockets)

$$B = I_{sp} V_0 (1 - \phi_e) \frac{C_L}{C_D}$$ ; $R \equiv$ range, $V_0$ = velocity.

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For transatmospheric flight

$$\frac{dm}{m} = \left\{ \frac{d\left(\frac{V_0^2}{2}\right) + g\,dr}{g_0 I_{sp} V_0 \left(1 - \frac{(D+D_e)}{\tau}\right)} \right\} \quad (1)$$

Note that $g_0 I_{sp} V_0$ is a measure of the efficiency of converting chem. enthalpy stored in the propellants into thrust ~~pow~~ power.

- To integrate the above eq$^n$, we want to make a few assumption
  - limit ourselves to LEO

$$\frac{r}{r_0} = 1 + \varepsilon \quad \& \quad \varepsilon \ll 1$$

& we showed earlier that

$$OE \simeq KE \quad (\text{neglect } PE)$$

$$OE = \frac{1}{2} v_{orb}^2 = \frac{1}{2} g_0 \frac{r_0^2}{r}$$

$$= \frac{1}{2} g_0 r_0 \left(\frac{1}{1+\varepsilon}\right)$$

$$\simeq \frac{1}{2} g_0 r_0$$

Since $\frac{r}{r_0} = 1 + \varepsilon$, $\Rightarrow$ neglect P.E in $\frac{dm}{m}$ eq$^n$ & can integrate assuming the part of the denominator $g_0 I_{sp}\left(1 - \left(\frac{D+D_e}{\tau}\right)\right) = \text{¢}$

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$$\Pi_f = 1 - \exp\left\{\frac{-(g_0 \gamma_0)^{1/2}}{g_0 I_{sp}\left(1 - \frac{D+D_e}{\tau}\right)}\right\}$$

For a pure rocket.

$$\Gamma_m = \left(1 - \Pi_f - \Pi_e\right)^{-1}$$

$$\Gamma_m = \left[\exp\left\{\frac{-(g_0 \gamma_0)^{1/2}}{g_0 I_{sp}\left(1 - \frac{D+D_e}{\tau}\right)}\right\} - \Pi_e\right]^{-1}$$

$$= \left[\exp\left(-\Delta\right) - \Pi_e\right]^{-1}$$

where $\Delta = \dfrac{(g_0 \gamma_0)^{1/2}}{g_0 I_{sp}\left(1 - \frac{D+D_e}{\tau}\right)}$

For a 2 stage rocket, we can show vehicle is optimized when stages are ~~set~~ self-similar $\left(\Delta_1 = \Delta_2,\ \Pi_{e_1} = \Pi_{e_2}\right)$

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Now look @ stage B

$$\Gamma_m = \Gamma_{m_1} \cdot \Gamma_{m_2}$$

$$\Gamma_{m_1} = \left[ \exp(-\alpha \Delta_1) - \Pi_{e_1} \right]^{-1}$$

$$\Gamma_{m_2} = \left[ \exp\left(-(1-\sqrt{\alpha})\Delta_2\right) - \Pi_{e_2} \right]^{-1}$$

Note that $\alpha$ is square-rooted in rocket stage because $\Delta$ scales with $(g_0 r_0)^{1/2}$ whereas in air breathers, $\Delta$ scales with $g_0 r_0$

For hypersonic TSTO w/ airbreather & rocket,

$$\Gamma_m = \left\{ \left[ \exp(-\alpha \Delta_1) - \Pi_{e_1} \right] \left[ \exp\left(-(1-\sqrt{\alpha})\Delta_2\right) - \Pi_{e_2} \right] \right\}^{-1}$$

$$\Delta_1 = f(\eta_0 H_{vf} \cdots)$$

$$\Delta_2 = f(I_{sp}, V)$$

For optimal performance

$$\Delta_1 \neq \Delta_2$$

$$\Pi_{e_1} \neq \Pi_{e_2}$$

Energy split, not symmetric, 1st stage does more work. (airbreathing engine)

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